

Regime-Aware Cross-Asset Signal Framework

Integrating Multi-Factor Analysis with LLM-Driven Scenario Generation — Methodology Evolution Log

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Purpose: A traceable record of how the project's methodology was reasoned through and refined — initial idea → critique → refined design → locked design — with consolidated formulae, data infrastructure, reproducibility protocol, literature anchors, and per-module next steps. This is a working methodology log to seed the Final Report and Learning Journal, not the final report itself.

How to read this document. Each module is presented as an *evolving process*. Grey blocks are the **initial idea**, orange blocks the **critique or correction**, blue blocks the **refinement**. The **green block states the LOCKED final design**.

Amber boxes give **next steps**. Points needing your supervisor's ruling are flagged inline like this: **Supervisor input needed** — never as a separate section.

Part I — Conceptual Foundations Established

Before any design choices, the project rests on a set of machine-learning fundamentals that were worked through from first principles. These are recorded here because every later method choice refers back to them.

1. Traditional algorithms vs. AI algorithms

A traditional algorithm is *specified*: a human writes the rules, and intelligence lives in the programmer's head (e.g. a hand-weighted hotel scorer, or a rule-based factor screen). An AI algorithm is *learned*: the programmer specifies an architecture and an objective, and the rules — the parameters — emerge from optimisation on data. The practical consequences distinguish them more than "better output": traditional code is deterministic, brittle outside its specification, fully explainable, and fails loudly; AI is probabilistic, generalises to unseen inputs, is largely opaque, and fails quietly and confidently. The mathematical base shifts from logic and proof to statistics and continuous optimisation — closer to econometrics than to classical programming.

The one-line distinction carried forward: traditional algorithms encode *how to do something*; AI algorithms encode *examples of something done* and infer the how. The project deliberately blends both — classical methods where the "how" is well-defined and auditable, learned methods where it is implicit.

2. Supervised vs. unsupervised learning

Supervised learning trains on examples that include the correct answer (a label) and learns to reproduce it on new inputs — regression (numeric target) and classification (categorical target). **Unsupervised** learning receives data with no answer key and finds structure on its own — clustering, dimensionality reduction, anomaly detection. Two hybrids matter for this project: **self-supervised** learning (the basis of modern LLMs — labels are generated from the data itself by hiding parts of it), and **reinforcement learning** (learning from reward signals). The OLS regression already familiar from prior coursework is, formally, a supervised method; matrix factorisation for recommenders is also supervised (labels are observed ratings).

Direct relevance to the project's framing of "the most important price-driving factor." The phrase pulls in two directions, and the project answers both:

- **SUPERVISED** Which observable, named factor best explains/predicts returns? — factor regression; coefficient magnitudes and significance give importance.
- **UNSUPERVISED** What hidden dimensions drive co-movement? — PCA discovers latent factors without naming them.

The regime layer is unsupervised (no ground-truth label exists for "which regime was the market in on a given day"); factor attribution within a regime is supervised; the LLM layer uses a pre-trained (self-supervised) model as a feature extractor. **Three inputs, three jobs, one combined output** — they are stacked, not competing.

3. The learning mechanism, made concrete

To demystify "it learns", the mechanism was worked through on matrix factorisation — the simplest algorithm that discovers latent structure. Every user and item is a short vector of latent traits; the predicted affinity is their dot product; the vectors begin as random numbers and are nudged toward values that minimise prediction error, repeated over the data. **The slots in the vectors are not assigned meaning in advance — meaning emerges from compression**, because forcing the data through a few dimensions makes the model find the few axes that best explain variation. This is precisely the intuition behind PCA latent factors and behind regime discovery later in the project.

PREDICTION (DOT PRODUCT OF LATENT VECTORS)

$$\hat{r}_{ui} = p_u \cdot q_i = \sum_f p_{uf} q_{if}$$

ERROR AND GRADIENT-DESCENT UPDATE (α = LEARNING RATE, λ = REGULARISATION)

$$e_{ui} = r_{ui} - \hat{r}_{ui} \\ p_u \leftarrow p_u + \alpha (e_{ui} \cdot q_i - \lambda p_u) \quad ; \quad q_i \leftarrow q_i + \alpha (e_{ui} p_u - \lambda q_i)$$

Interpretation: scale each correction by how wrong we were ($\times$ error); move each vector in the direction of the other (the slots where the partner is strong matter most); take small steps ($\times \alpha$) so no single observation dominates; the λ term keeps vectors from growing unbounded (overfitting control). The learning rate governs *how aggressively each data point nudges the parameters* — it is independent of sample size. This same "predict $\rightarrow$ measure error $\rightarrow$ nudge" loop, scaled up, trains every model in the project.

Part II — The Framework and Its Three Problems

The deliverable is a hybrid regime-and-flow detection framework for cross-asset signal extraction, combining classical multi-factor analysis with an LLM layer (retrieval-augmented). It investigates three contested problems in empirical asset pricing and market microstructure:

- **Problem 1 — Conditional safe-haven behaviour / inversion.** Whether safe-haven assets (notably gold) merely attenuate under stress or actually *invert* to behave as risk assets under acute liquidity conditions.
- **Problem 2 — Capital rotation within thematic chains.** Whether sequential capital flow along supply chains (e.g. the semiconductor cascade) is statistically detectable, and whether rupture events that break expected sequencing can be identified without overfitting.
- **Problem 3 — The appropriate role of LLMs in a financial pipeline.** Whether a principled middle position is defensible: LLMs as structured feature extractors and reasoning agents over unstructured text, with numerical inference left to classical methods.

Part III — Methodological Evolution (Module by Module)

Module A • Pipeline architecture and ordering

INITIAL IDEA

Run RAG first to find signals from text, then collect factor data, then asset prices; treat supervised, unsupervised and LLM as a sequence where unsupervised "finds what is left over" after supervised and the LLM.

CRITIQUE

RAG cannot run first — it must know *which* assets and *which* dates to retrieve against, so it depends on the price/asset spine. The three methods are not a sequence competing on one job; they operate on different inputs and answer different questions.

REFINEMENT

Build the data spine first (prices, macro, text). Then apply three parallel methods: unsupervised on macro data (regimes + latent factors), supervised on returns (named-factor importance, conditional on regime), LLM on text (features). They converge into one regime-aware signal.

LOCKED DESIGN

Data spine precedes methods. Collect (i) asset prices, (ii) macro/factor series, (iii) text corpus — in that priority. Then: unsupervised characterises the state of the world (regimes); supervised measures factor-return relationships given that state; the LLM reads text to add features and reasoning. Three inputs, three jobs, one combined output — stacked, not sequential rivals.

Module B · Scope strategy — the "engine" reframe

INITIAL IDEA

Study three full rotation systems (metals, semiconductors, macro risk), a dynamic regime-discovery engine, a six-variable inversion study, and a four-source ten-year text corpus.

CRITIQUE

That is roughly five separate research projects, each capable of filling thirteen weeks. A general-purpose engine that ingests all datasets is *harder* to build than one focused study — generality is expensive, not cheap.

REFINEMENT

Reframe as engines (rotation detector, regime engine, inversion engine), but make them *general in architecture* and *validated narrowly* in the MVP: prove on one dataset, architect so others plug in via config + data adapter.

LOCKED DESIGN

Build each engine general in architecture but validate on one dataset in the MVP. The deferred datasets become roadmap items in the business plan, not work done in 13 weeks. The defensive sentence for the whole project is: "engine general by design, validation narrow by discipline."

NEXT STEPS — MODULE B

1. Write the one-line scope-defence sentence into the project charter and the Final Report introduction.
2. Define the plug-in contract (config schema + data-adapter interface) so a second chain could be added without code rewrites.

SUPERVISOR INPUT NEEDED: confirm that an MVP validated deeply on a single rotation chain (with others as documented roadmap) satisfies the course's deliverable expectations, rather than breadth across all chains.

Module C · Regime engine — discovery vs. classification, and graded crises

INITIAL IDEA

A "dynamic" regime system that, whenever unsupervised learning discovers a new latent trend, re-selects and re-investigates the matching historical regime in real time.

CRITIQUE

Continuous re-discovery is a circular feedback loop: regime definitions never sit still long enough to validate, and outputs are not comparable over time — it cannot be sold or graded. The conflation was between two different meanings of "dynamic".

REFINEMENT

Separate the two. *Discovery* (clustering that defines regimes) runs **offline**, periodically, is validated, then frozen. *Classification* (labelling today against frozen regimes) runs **online**, daily, cheaply — this is the sellable feed. Retrain discovery on a schedule, not on every wiggle.

Two further refinements were folded in. First, run regime discovery on *macro variables* (long, clean histories) rather than on the asset basket (many semiconductor proxies did not exist 10–15 years ago); the asset study then runs conditional on the macro regime, decoupling the two histories. Second, **"crisis" is a graded spectrum, not three systemic giants**: alongside 2008 / 2020 / 2022 sit 1987, 1994, 1998 (LTCM), 2000, 2011, 2015–16, 2018Q4, and narrow asset-specific drawdowns such as the February 2026 silver episode — sharp, sector-local, weeks-to-months to recover. The engine must therefore detect stress across multiple severities and breadths, not as a binary.

LOCKED DESIGN

Offline discovery → freeze → online classification. Cluster on macro variables for long history; study assets conditional on the macro regime. Detect crises as a graded continuum of severity and breadth (systemic to asset-specific), not a binary. "Dynamic" means real-time classification against frozen definitions — never real-time re-discovery.

NEXT STEPS — MODULE C

1. Select the macro variable panel for clustering (rates, spreads, volatility, M2, funding-stress proxies).
2. Choose the clustering method (k-means / Gaussian mixture / Markov regime-switching) and fix a random seed for reproducible regimes.
3. Compile the graded episode list (systemic + sector-specific) with start/end dates to serve as the backtest set and as validation for the discovered regimes.
4. Set the retraining cadence for offline discovery (e.g. quarterly).

SUPERVISOR INPUT NEEDED: agree the graded crisis-episode list and the number/granularity of regimes; confirm the macro-clustering-then-asset-conditioning split is methodologically acceptable for Problem 1 and Problem 2.

Module D • Inversion engine (Problem 1)

INITIAL IDEA

Variables: US bond yield, gold, tech-crash illiquidity effect on gold, oil → USD pressure on gold, M2 liquidation status, overnight LIBOR.

CRITIQUE

USD LIBOR is discontinued; using it is anachronistic. Nominal yield is weaker than the real-yield link for gold. The set needed a single explicit liquidity-stress measure.

REFINEMENT

Replace LIBOR with SOFR-based stress; use *real* yields as the cleanest gold driver; add an explicit liquidity-stress proxy.

LOCKED DESIGN

Six variables, all with long FRED histories, measured regime-conditionally: US real yields; gold; a liquidity-stress proxy (SOFR–EFFR spread and/or the MOVE index); oil; the US Dollar Index (DXY); and M2. This is the most code-ready component. The inversion question is operationalised as: does the gold–risk correlation flip sign conditional on the liquidity-stress regime?

NEXT STEPS — MODULE D

1. Pull the six series; construct the SOFR–EFFR spread and align frequencies.
2. Specify the regime-conditional regression (below) and a first empirical look at sign-flips during stress windows.
3. Document the SOFR splice for pre-2018 history (see Module on data constraints).

SUPERVISOR INPUT NEEDED: approve the pre-2018 funding-stress proxy splice (historical TED spread / LIBOR-OIS joined to SOFR–EFFR) so the long backtest is defensible.

REGIME-CONDITIONAL FACTOR REGRESSION (SUPERVISED CORE)

$$r_{i,t} = \alpha_s + \sum_k \beta_{k,s} F_{k,t} + \varepsilon_{i,t} \quad \text{for regime state } s$$

Importance of factor k is read from $\beta_{k,s}$ and its significance, *within each regime* s . Inversion appears as a sign change in the gold-on-risk coefficient across regimes.

Module E • Rotation engine (Problem 2)

INITIAL IDEA

Three chains: metals (gold/silver/"bronze"/oil/rare earths), the full semiconductor cascade (GPU/CPU/DRAM/HBM/optical/packaging/data-centre power/nuclear), and macro risk rotation (equities/treasuries/gold).

CRITIQUE

Three chains is unmanageable. Most semiconductor links are not directly investable (HBM, packaging, optical are product categories, not instruments) and must be noisily proxied. "Bronze" is not a traded metal — the intended bellwether is copper. Rare earths are barely investable (one thin ETF). Gold and oil already do work in the inversion and macro layers.

REFINEMENT

One deep chain + one always-on backbone + architecture for extension. Drop the standalone metals chain (gold/oil retained elsewhere; copper noted for any future revival).

LOCKED DESIGN

Primary deep-dive: the semiconductor chain, proxied by the 4–5 cleanest links (design → memory → packaging → data-centre power), with explicit acknowledgement that finer links are out of MVP scope and that single-name proxies inject idiosyncratic noise. **Always-on backbone:** the macro risk rotation (equities ↔ treasuries ↔ gold), which doubles as a regime signal. Metals/rare-earths deferred to the roadmap.

NEXT STEPS — MODULE E

1. Finalise the proxy basket for each retained link and record each instrument's history start (proxies have unequal, often short, histories).
2. Specify the lead-lag estimation method and a rupture-detection rule (a substitution technology bypassing a previously load-bearing link).

SUPERVISOR INPUT NEEDED: sanity-check the semiconductor proxy mapping and confirm that proxy noise is an acceptable, documented MVP limitation rather than a fatal flaw.

Module F · Historical analog and forecasting — the dual-kernel weighting

INITIAL IDEA

Use unsupervised learning to find a similar historical regime and treat the resemblance as a future prediction; weight recent regimes more, distant ones less, and give a "perfect" match ($1-2\sigma$) the highest weight to determine the driving force and future direction.

CRITIQUE

Resemblance is not prophecy — markets are non-stationary and crises are few, so confident point prediction overfits. A "perfect match within 1σ " across ~20 raw variables essentially never occurs (curse of dimensionality). The scheme is hyperparameter-heavy and will overfit if tuned to make the backtest look good.

REFINEMENT

Output a *conditional distribution with uncertainty*, not a point forecast. Formalise the weighting as locally-weighted (kernel) learning on two axes — time-decay and feature-similarity — and measure similarity in **PCA-reduced space**, not raw variables. Fix hyperparameters by economic reasoning and light cross-validation; document them.

LOCKED DESIGN

Forecasts are weighted empirical conditional distributions over historical analogs. Each historical period's weight combines an exponential recency kernel and a Gaussian similarity kernel computed in PCA space. Because the stated intent is for a near-perfect match to dominate regardless of age, the blend is similarity-dominant (tight σ so only true analogs get weight; gentle recency decay breaks ties among them) — an explicit, documented choice.

RECENCY KERNEL ($T = \text{NOW}$, $t = \text{HISTORICAL DATE}$, $\lambda = \text{DECAY RATE}$)

$$w_t^{\text{time}} = \exp(-\lambda (T - t))$$

SIMILARITY KERNEL IN PCA SPACE ($z = \text{PCA PROJECTION OF THE STATE}$, $\sigma = \text{BANDWIDTH}$)

$$w_t^{\text{sim}} = \exp(-\|z_t - z_{\text{now}}\|^2 / (2\sigma^2))$$

COMBINED WEIGHT, AND THE RESULTING FORWARD-RETURN FORECAST (HORIZON h)

$$w_t = w_t^{\text{time}} \cdot w_t^{\text{sim}} \quad (\text{similarity-dominant via small } \sigma)$$

$$\hat{E}[r_{t+h}] = (\sum_t w_t r_{t+h,t}) / (\sum_t w_t) \quad ; \quad P(\text{up}) = (\sum_t w_t \cdot 1[r_{t+h,t} > 0]) / (\sum_t w_t)$$

NEXT STEPS — MODULE F

1. Decide the PCA component count used for the similarity metric and the horizon(s) h .
2. Set λ and σ by reasoning + cross-validation; record values in the config; report sensitivity.
3. Report forecasts as distributions (probability of up-move + range), never as single numbers.

SUPERVISOR INPUT NEEDED: confirm the similarity-dominant blend (vs. a balanced multiplicative blend) matches the intended research claim, and agree an overfitting-control protocol given the small number of true analogs.

Module G · LLM / RAG — from "commentator" to "analyst with a calculator" (Problem 3)

INITIAL IDEA

Use the LLM's theoretical reasoning to deduce the future return and direction directly from the situation.

CRITIQUE

An LLM asked to emit a return is a black-box predictor producing fluent, unauditable confabulation — exactly the failure mode Problem 3 warns against. But restricting the LLM to mere caveats was too timid for the intended product.

REFINEMENT

Promote the LLM to *analyst*, keep one iron rule. The LLM may hypothesise, investigate, correlate news, and write the verdict — but it never fabricates a number; it **commissions** every figure from the quantitative engine (tool-grounded / agentic). It notices a difference from the analog, forms a testable hypothesis, triggers a fresh computation, reads the real result, checks corroboration across independent news sources, and only then synthesises the verdict.

LOCKED DESIGN

The LLM is the analyst; the quantitative engine is the calculator it is forbidden from making up figures without. Iron rule: every quantitative claim in the verdict traces to a computation, never to the model's imagination. The LLM's value-add is hypothesis generation, cross-news corroboration, and narrative synthesis — not number generation. The target verdict maps to grounded sources clause by clause:

- *"the main price-driving force we calculated was ..."* → supervised regime-conditional regression coefficient.
- *"for the next ... days the momentum is ongoing ..."* → analog conditional distribution + regime-persistence statistics.
- *"but because we have seen ... factors that could stop it, also proven in past history ..."* → LLM hypothesis, then engine validation against historical episodes.
- *"these following factors can be a reverse sign ..."* → LLM-flagged, quant-corroborated watchlist; confidence set by news corroboration strength × historical support.

NEXT STEPS — MODULE G

1. Start the corpus with the two free, structured, high-signal sources only: SEC EDGAR filings + FOMC text. Defer proprietary IB research (not legally obtainable); handle shock/event news via an event API.
2. Design the tool interface: the set of computations the LLM is allowed to commission (factor regression, correlation, event-window stats, corroboration scoring).
3. Define the verdict template and the rule that blocks any number not returned by a tool call.
4. Track news corroboration via embedding similarity + co-occurrence + semantic judgement.

SUPERVISOR INPUT NEEDED: endorse the "analyst-with-a-calculator" architecture as the principled middle position for Problem 3, and agree how incremental signal over a pure-quant baseline will be measured (and what counts as ground truth when ground truth is itself contested).

Module H · Crisis early-warning — leading-indicator probability gauge

INITIAL IDEA

Detect crash/"bomb" events; race to react as fast as possible.

CRITIQUE (SELF-IDENTIFIED)

Crises cannot be predicted to the day, and fighting quant systems on anomaly-detection *speed* (colocation, microsecond latency) is unwinnable. Reacting late means the move is already gone.

REFINEMENT

Build a leading-indicator probability gauge from slower-moving structural signs that *lead* by weeks-to-months, output a probability with a threshold, and advise position reduction above it. Because leading indicators lead, a daily (close-based) update is fast enough — this sidesteps the latency war entirely and answers the "react too slowly" worry.

LOCKED DESIGN

A composite stress-probability model (interpretable, e.g. logistic) over leading indicators — yield-curve inversion, credit-spread widening, funding stress, volatility-term-structure backwardation, MOVE, breadth deterioration, and the regime classifier's own drift toward the crisis cluster. Output P(crisis within N weeks) with a de-risk threshold. Framed as a risk-management overlay, NOT a market-timing oracle, because leading indicators carry false positives. Daily update cadence is sufficient and deliberate.

CRISIS PROBABILITY (LOGISTIC; X = LEADING-INDICATOR VECTOR)

$$P(\text{crisis}_{t \rightarrow t+N}) = 1 / (1 + \exp(-(\beta_0 + \sum_k \beta_k x_{k,t})))$$

NEXT STEPS — MODULE H

1. Assemble the leading-indicator panel; verify each genuinely leads (not coincides).
2. Label historical crisis windows (graded, per Module C) as the supervised target.
3. Fit the gauge, evaluate false-positive rate honestly, and set the de-risk threshold.

SUPERVISOR INPUT NEEDED: agree the labelling of crisis windows and an acceptable trade-off between early warning and false positives for the threshold.

Module I · The dual analytical layers (settled and carried forward)

PCA = Principal Component Analysis: the canonical unsupervised dimensionality-reduction method, which finds the combined directions of maximum variance in the data; the first component of equity returns is typically "the market", later components capture style/sector structure. PCA is **fit offline (discovery), frozen, and applied online (classification)** — the same pattern as the regime model. It plays two roles: a cleaning step before clustering, and the discovered-factor product layer.

LOCKED DESIGN — THE RESOLUTION TO "WHICH FACTOR APPROACH"

Do both, layered. The named-factor, regime-conditional analysis is the validated academic core (rigorous, auditable) for the Final Report and supervisor. The PCA discovered-factor layer is the self-evolving product differentiator (it updates as data arrives and can surface genuinely new drivers) for the business plan and retail-facing product. Same data, two layers, two audiences — the project does not choose between them, it stacks them.

Why discovered-factor is the more "self-evolving" product: as new data arrives and PCA is refit, the latent factors update, so the system can flag "a new driver is now active". The named-factor route is more defensible but static. For a daily retail product, the evolving narrative is the differentiator; for the report, the named/defensible core is the rigour.

Part IV — Consolidated Statistical Formulae

All formulae referenced above, gathered for the report's methodology appendix.

Purpose	Formula
Latent-affinity prediction (learning-mechanism illustration)	$\hat{r}_{ui} = p_u \cdot q_i = \sum_f p_{uf} q_{if}$
Gradient-descent parameter update	$e_{ui} = r_{ui} - \hat{r}_{ui}; \quad p_u \leftarrow p_u + \alpha(e_{ui} q_i - \lambda p_u); \quad q_i \leftarrow q_i + \alpha(e_{ui} p_u - \lambda q_i)$
Regime-conditional factor attribution (supervised core)	$r_{i,t} = \alpha_s + \sum_k \beta_{k,s} F_{k,t} + \varepsilon_{i,t} \quad (\text{per regime } s)$
PCA latent factors (unsupervised discovery)	first component $w_1 = \operatorname{argmax}_{\ w\ =1} \operatorname{Var}(Xw)$; components are eigenvectors of the covariance matrix Σ (or via SVD of X)
Recency kernel	$w_t^{\text{time}} = \exp(-\lambda(T-t))$
Similarity kernel (in PCA space)	$w_t^{\text{sim}} = \exp(-\ z_t - z_{\text{now}}\ ^2 / 2\sigma^2)$
Combined analog weight + forecast	$w_t = w_t^{\text{time}} \cdot w_t^{\text{sim}}; \quad \hat{E}[r_{t+h}] = \sum w_t r_{t+h,t} / \sum w_t; \quad P(\text{up}) = \sum w_t \cdot 1[r_{t+h,t} > 0] / \sum w_t$
Crisis early-warning probability	$P(\text{crisis}_{t \rightarrow t+N}) = 1 / (1 + \exp(-(\beta_0 + \sum_k \beta_k x_{k,t})))$

Part V — Data Infrastructure

Architecture principle: **boringly simple and fully reproducible** — no database server, no orchestration pipeline. Config-driven download scripts, a raw cache, Parquet files on disk queried with DuckDB (SQLite as the simpler fallback), local vector store (ChromaDB or FAISS) for RAG.

SOURCES, EARLIEST DATA, AND BINDING CONSTRAINTS

Source	Earliest data	Notes & constraints
yfinance (Yahoo)	Indices ~1927; many US stocks floor 1962-01-02; ETFs from inception (SPY 1993, GLD 2004, SLV 2006, USO 2006, CPER 2011)	Quality degrades far back; not survivorship-safe. Use as free, reproducible fallback — not system of record.
FRED	Series-specific: M2 1959; nominal 10Y (DGS10) 1962; real 10Y (DFII10) 2003 ; SOFR 2018-04 ; EFFR 1954; VIX 1990; broad USD index 2006	Workhorse for all macro/inversion series. SOFR (2018) and real yields (2003) are the binding constraints on long backtests.
SEC EDGAR	~1993–1994	Electronic filings only from then. The 10-year filing window (2016–2026) is fully covered.
GDELT	Events 1979; entity/theme richness (GKG) 2013 (2.0 from Feb 2015)	Use for entity/theme-tracked news + rupture/shock detection. Pre-2013 has only thin event data.
FOMC text		Clean, high-signal policy corpus, free.

	Statements 1994; transcripts on a 5-year lag	
Refinitiv / LSEG	Broadly 1970s; some series older; includes delisted securities	Deepest accessible history; survivorship-friendlier. Library renamed: use lseg-data (LSEG Data Library v2); the old eikon API is deprecated. Desktop session via Workspace/Eikon authenticates without a config file.

On "as far back as the bourse started": no accessible source reaches exchange inception, and the project does not need it — it needs enough history to *contain its graded crisis episodes*. Coverage to the 1990s (or 1970s via Refinitiv) is more than sufficient. Design each study to respect its shortest necessary series or splice a documented proxy.

NEXT STEPS — DATA

1. Use Refinitiv/LSEG as primary for prices (long history + delisted names → fixes survivorship bias); FRED for macro; yfinance as free fallback; EDGAR + FOMC + GDELT for text/events.
2. Build the config-driven download scripts with a raw cache; write to Parquet; stand up DuckDB.
3. Record each series' start date and any splice in the data dictionary.

SUPERVISOR INPUT NEEDED: note for the business plan that Refinitiv data generally *cannot be redistributed* under university licensing — a commercial product would need a proper data licence. Confirm this is acceptable for the academic MVP.

Part VI — Reproducibility Discipline (with concrete steps)

Practice	Concrete steps to fulfil it
Configuration-driven setup	One <code>config.yaml</code> at the root listing every ticker, FRED series ID, date range, and hyperparameter (λ , σ , PCA components, regime count). Every script reads from it; nothing is hard-coded. Changing scope = editing one file.
Raw-cache layer	Wrap every API call in check-then-fetch-then-save: if a date-stamped raw file exists, load it; else call the API and immediately save the untouched response. Treat <code>raw/</code> as write-once.
Layer separation	Three folders — <code>raw/</code> (untouched), <code>processed/</code> (cleaned, aligned), <code>outputs/</code> (results, figures). Data flows one way only; never patch a processed file by hand — regenerate it.
Determinism	Pass an explicit fixed seed to every stochastic step (clustering init, splits) and record it in config. Verify by running the pipeline twice and confirming identical regimes.
Environment pinning	<code>pip freeze > requirements.txt</code> ; note the Python version. Rebuild into a fresh virtual environment so versions match exactly.
Point-in-time correctness	Never use a value before it was knowable: filter to timestamp $\leq$ decision date. For revised series (M2 etc.) use FRED's archival ALFRED vintages, not latest-revision. Route every backtest query through one "as-of" function so look-ahead cannot sneak in.
Single rebuild command	One top-level script/notebook that regenerates everything from a clean checkout + raw cache. Test by deleting <code>processed/</code> and <code>outputs/</code> and rerunning; any manual step that breaks it must be eliminated.
Data dictionary	A markdown table mapping each series to source, ID/ticker, frequency, units, earliest date, and quirks (e.g. "SOFR pre-2018 spliced from TED spread"). Add a row the moment any series is introduced.

Two finance-specific integrity issues to *document* rather than fully solve in the MVP: **survivorship bias** (a 10-year backtest on today's surviving tickers — mitigated by using Refinitiv's delisted-inclusive data) and **look-ahead bias** (addressed by point-in-time correctness above). Naming known biases explicitly is what makes the work defensible.

Part VII — Literature Anchor Map

Five clusters, mapped to the modules. Start from the foundational anchors, then chase citations forward. The text/LLM and alternative-data clusters move fast — anchor on foundations but search recent work explicitly.

Cluster	Search topics	Canonical anchors
Regime detection (Module C)	Markov regime-switching; hidden Markov models in markets; Gaussian-mixture / clustering of market regimes	Hamilton (1989) regime-switching
Safe-haven & inversion (Module D)	gold hedge vs. safe haven; flight to liquidity; dash for cash 2020; gold–real-rate–dollar link	Baur & Lucey; Baur & McDermott
Rotation & lead-lag (Module E)	economic links and predictable returns; industry/sector momentum; lead-lag in supply-chain returns	Cohen & Frazzini (supply-chain returns)
LLM / text in finance (Module G)	textual analysis in finance; FinBERT; retrieval-augmented generation	Loughran & McDonald (financial sentiment); Lewis et al. (RAG). Underpinning: Fama-French 3/5-factor; Chen-Roll-Ross APT
Financial data construction (Part V)	FRED-MD macro database; GDELT global events; nowcasting & real-time data revisions; alternative data in asset management	McCracken & Ng (FRED-MD); Leetaru & Schrodtt (GDELT)

SUPERVISOR INPUT / OPEN OFFER: the alternative-data and LLM-for-finance literature turns over quickly; a live search of recent work (post-foundational) is recommended so Problem 3 and the data-construction section cite current methods, not only canonical papers.

Closing note

The two ideas worth foregrounding to the supervisor as genuinely the student's own contribution are: **(1) the LLM-as-analyst-with-a-calculator architecture** (Module G — principled, auditable, the centrepiece of Problem 3) and **(2) the dual-kernel recency × similarity weighting in PCA space** (Module F). Both are defensible and slightly novel. The natural next concrete step is to turn this locked scope into the asset-universe document and the `config.yaml`, where Week 1's real deliverable lives.